

Question Paper

Exam Date & Time: 27-Dec-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
SEVENTH SEMESTER B.TECH. DEGREE MAKEUP EXAMINATIONS - DECEMBER 2025
BLOCK CHAIN TECHNOLOGY [CSE_4033]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Apply the concept of distributed record keeping to design a system that eliminates the need for a central bank in digital transactions. Demonstrate how blockchain can solve issues in traditional centralized ledgers like double-spending or single-point failure. (5)
- A)

Course Outcome: "CO1" / Cognitive Level: "Application"

- B) Compare traditional banking systems with Nakamoto's blockchain model in terms of fault tolerance and trust mechanisms. Examine the economic and security motivations behind the creation of blockchain-based cryptocurrencies. (5)

Course Outcome: "CO2" / Cognitive Level: "Analysis"

- 2) Apply the idea of consensus without identity to explain how Bitcoin reaches agreement among anonymous participants. (5)

A)

Course Outcome: "CO1" / Cognitive Level: "Application"

- B) Analyze the importance of atomic broadcasting and its significance. Compare the different cryptographic approaches used in cryptographic hash functions. (5)

Course Outcome: "CO2" / Cognitive Level: "Analysis"

- 3) Examine the structure of the Bitcoin blockchain with a neat diagram. (5)

A)

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- B) Compare Pay-to-PubKey-Hash (P2PKH) and Pay-to-Script-Hash (P2SH). (3)

Course Outcome: "CO5" / Cognitive Level: "Analysis"

- C) Analyze how gas mechanisms and computational limits collectively prevent infinite loops and denial-of-service vulnerabilities in blockchain execution environments. (2)

Course Outcome: "CO5" / Cognitive Level: "Analysis"

- 4) Apply the concept of mining pools to explain how collective mining and reward distribution function in Bitcoin. Use examples to show how pooling affects miner participation and network performance. (5)

A)

Course Outcome: "CO1" / Cognitive Level: "Application"

- B) Develop a model to illustrate how the evolution of Bitcoin mining hardware-from CPU to ASIC-has reshaped incentive structures, profitability, and network decentralization. (3)

Course Outcome: "CO3" / Cognitive Level: "Application"

- C) Given a user who wishes to make an anonymous payment using Bitcoin, apply the concept of mixing to plan a privacy-preserving transaction flow. Describe how coin-mixing or CoinJoin mechanisms can be used to obscure transaction links. (2)

Course Outcome: "CO4" / Cognitive Level: "Application"

- 5) Inspect how off-chain data (e.g., IP addresses, exchange KYC) helps link Bitcoin identities and analyze its implications for user privacy and anonymity. (5)

- A) Course Outcome: "CO4" / Cognitive Level: "Analysis"

- B) Develop a model for a secure multi-party lottery using Bitcoin transactions. (3)

Course Outcome: "CO3" / Cognitive Level: "Application"

- C) Outline and explain the three types of consensus that operate in Bitcoin. (2)

Course Outcome: "CO5" / Cognitive Level: "Understand"

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Event Name	Subject
058-MIT BLR-MAKEUP EXAM-2025-2026	CSE_4033:BLOCK CHAIN TECHNOLOGY

#	Course Outcome	Description	Marks
1	1	CO 1	15
2	2	CO 2	10
3	3	CO 3	11
4	4	CO 4	7
5	5	CO 5	7
Total			50

* **NA** - CO Not Selected

Event Name	Subject
058-MIT BLR-MAKEUP EXAM-2025-2026	CSE_4033:BLOCK CHAIN TECHNOLOGY

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	1	2(4%)
3	Application	6	23(46%)
4	Analysis	6	25(50%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		13	50 (100%)